

ZERO TRUST FRAMEWORK

INSTRUCTOR'S PROFILE

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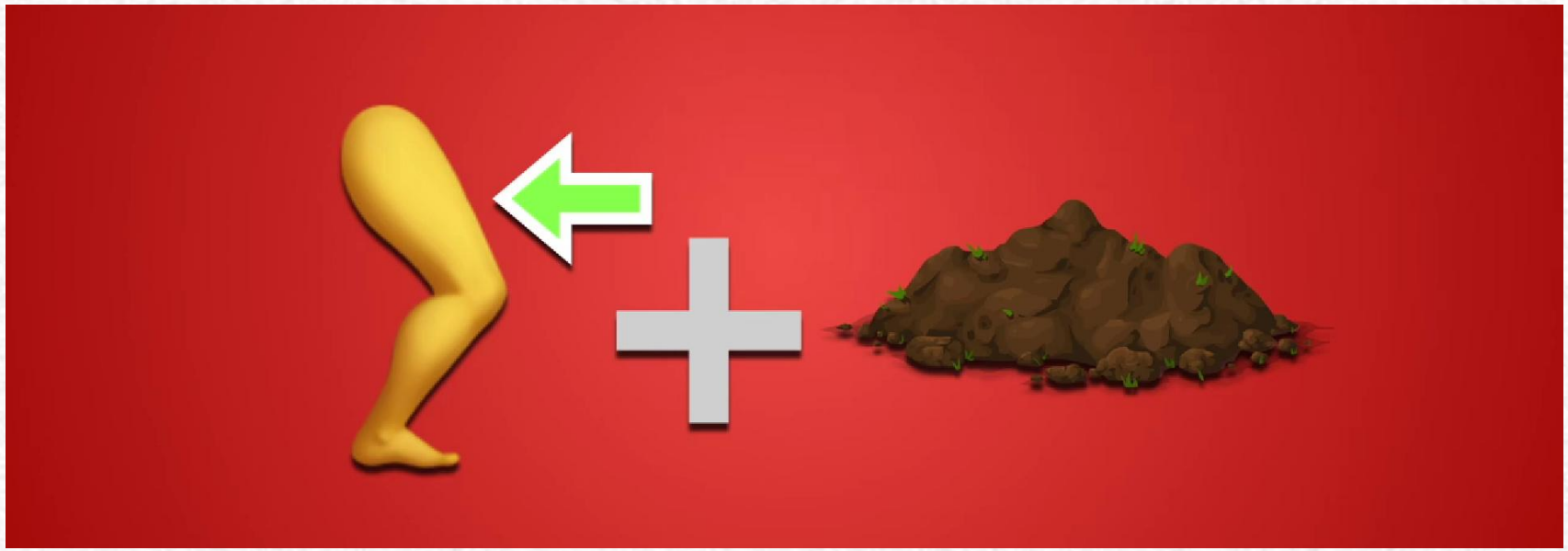
Education:
Philippine Military Academy "SALAKNIB" CL 2017

Certification:
- Cisco Certified Network Associate (02 July 2024)
- ISO Lead Auditor (20 March 2026)
- Cyber Threat Intelligence Management NCIII
- Data Analytics NCIII

Training/Exercises:
- AMBER MIST Cyber Exercise- Lithuania
- PH-AUS Defend Forward Cyber Exercise
- SABAK 2024-2026
- DAGITPA 2025

ICE BREAKER

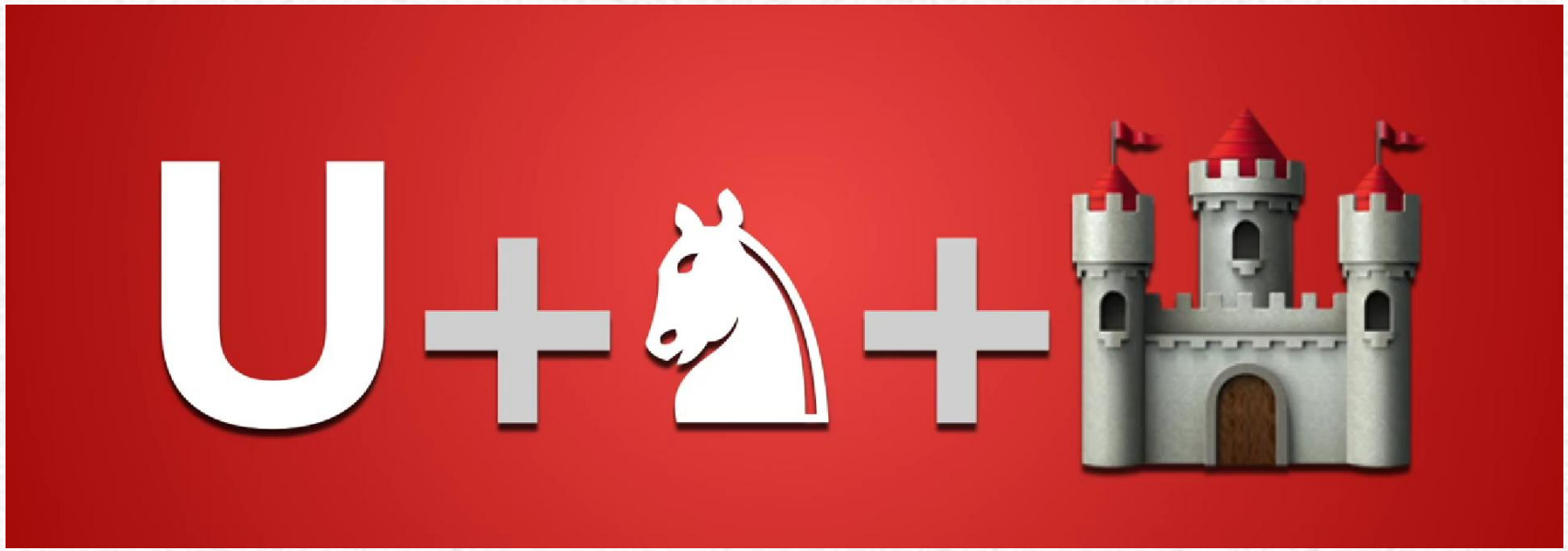
GUESS THE COUNTRY



THAILAND

ICE BREAKER

GUESS THE COUNTRY



UNITED KINGDOM

ICE BREAKER

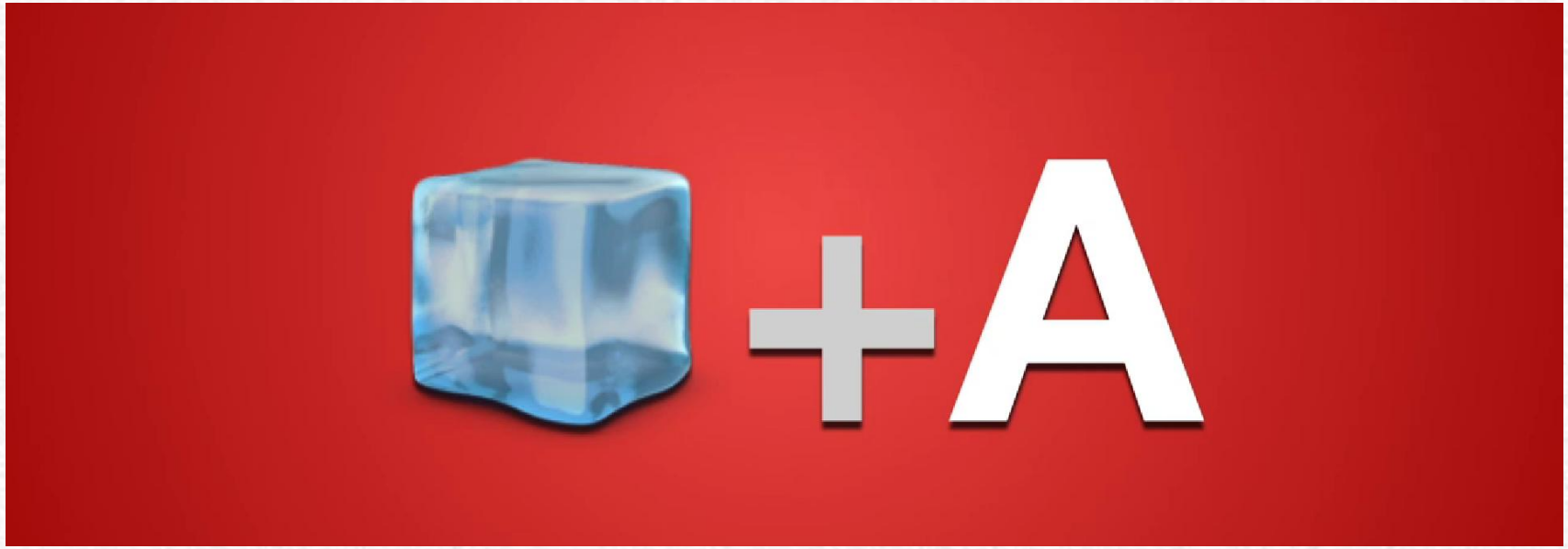
GUESS THE COUNTRY



GERMANY

ICE BREAKER

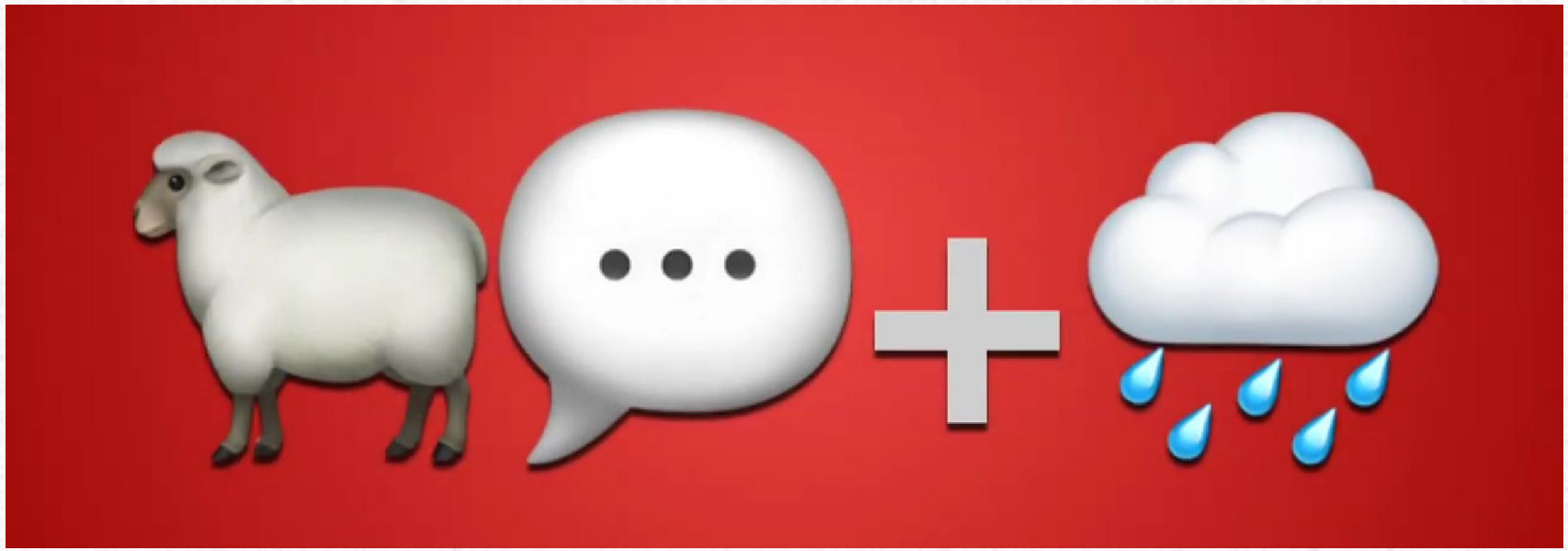
GUESS THE COUNTRY



CUBA

ICE BREAKER

GUESS THE COUNTRY



BAHRAIN

NEED

Studying and understanding the Zero Trust framework in order to shift defense from static, network-based perimeters to the continuous verification of every user, device, and access request, protecting resources, not network segments, of the Philippine Army.

TIME



ROADMAP

- **Diagnose** the following:
  - What is Zero Trust and Zero Trust Architecture (ZTA)
  - The seven tenets of Zero Trust
  - Logical components: PE, PA, PEP
  - Deployment approaches and variations
  - Threats associated with ZTA
  - Migrating to a Zero Trust Architecture
  - Applicability to the Philippine Army
- **Experiment:** check on learning
- **Reflect:** takeaways on the lesson
- **Evaluate:** Written Examination (Essay)

LEARNING OUTCOME

At the end of this subject, the students will be able to apply Zero Trust Framework in cyberspace during army operations in accordance with PAM 6-0102-4. (C3)

REFERENCE

- PAM 6-00 (Command and Control Communications and Cyber Systems)
- PAM 6-0102 – Signal Operations
- PAM 3-00 (Army Operations)
- PAM 6-0102-04 Cyberspace Operation
- NIST Special Publication 800-207 — Zero Trust Architecture

SAFETY PRECAUTION



The nearest person on the fire extinguisher will extinguish and control the fire. The facilitator will be the last person to leave the classroom. Consolidation point will be Infront of Yan Hall.

SAFETY PRECAUTION



In case of earthquake: **“DROP, COVER AND HOLD ON.”** and the emergency evacuation plan before the start of the training and ensure that the learner understands.



SAFETY PRECAUTION



Learner to be observant of broken electrical wires.

“SAFETY in mind, leave accidents behind.”



★ ★ — ARMY 2040: WORLD-CLASS. MULTI-MISSION READY. CROSS-DOMAIN CAPABLE. — ★ ★

HEAT INDEX SAFETY



Due to the high heat index being experienced nationwide, the Office of the Chief Surgeon reiterates the importance of observing heat safety measures during training and outdoor activities.
All personnel are enjoined to strictly follow the following guidelines.

	SAFETY MEASURE	GUIDELINES
1	STAY HYDRATED	• Drink plenty of water even if not thirsty. • Avoid sugary, caffeinated, and alcoholic drinks. • Hydrate before, during, and after activities to maintain optimal performance and alertness.
2	SEEK PROPER SHELTER	• Take frequent breaks under shaded or covered areas. • Use hats, caps, or umbrellas to protect from direct sun exposure.
3	PACE YOURSELF FOR THE DAY	• Adjust your pace with rest cycles in mind. • Slow down and avoid overexertion. • Build up your activity gradually while closely monitoring your body's response.
4	WEAR COOL, LIGHTWEIGHT CLOTHING	• Lightweight, breathable, moisture-wicking clothing helps regulate body temperature. • Light-colored clothing reflects sunlight and helps keep cool.
5	MONITOR FELLOWS (BUDDY SYSTEM)	• Observe one another for signs of heat exhaustion or heat stroke such as dizziness, confusion, excessive sweating, and weakness. • Report early. Report and assist anyone showing symptoms.
6	IMMEDIATE ACTION IF HEAT ILLNESS OCCURS	• Stop activity immediately. • Move to a cool, shaded area. • Provide water or oral rehydration (if not nauseated). • Provide first aid and seek medical attention.
7	ASSOCIATE COMMANDER SUPERVISION	• Ensure leaders are heat-index aware, maintain appropriate work-rest cycles, personnel with chronic medical conditions, and those under medication or vitamin supplements. • Leaders shall disseminate and enforce health and heat safety protocols.


THINK
OUR AIM . . .
NO ACCIDENTS

CLASSROOM RULES













Source of Army Standards

WHAT IS ZERO TRUST?

Zero Trust (ZT) is a set of cybersecurity paradigms that moves defenses from static, network-based perimeters to focus on users, assets, and resources. No implicit trust is granted based on physical or network location or on asset ownership.

ZT

The concept

Minimize uncertainty in enforcing accurate, least-privilege, per-request access decisions in a network viewed as compromised.

ZTA

The plan

An enterprise cybersecurity plan built on zero trust concepts — component relationships, workflow planning, and access policies.

ZT ENTERPRISE

The result

The network infrastructure (physical and virtual) and operational policies in place as a product of the ZTA plan.

Source of Army Standards

WHAT IS A ZERO TRUST FRAMEWORK?

Trust is never granted implicitly — it must be continually evaluated.

Every access request — by any user, device, or service, from any location — is authenticated, authorized, and continuously validated before and during access to a resource.

1 ASSUME BREACH

Operate as if an attacker is already present inside the network.

2 VERIFY EXPLICITLY

Authenticate and authorize every subject and device, per session.

3 LEAST PRIVILEGE

Grant only the minimum access needed — and shrink implicit trust zones.

Source of Army Standards

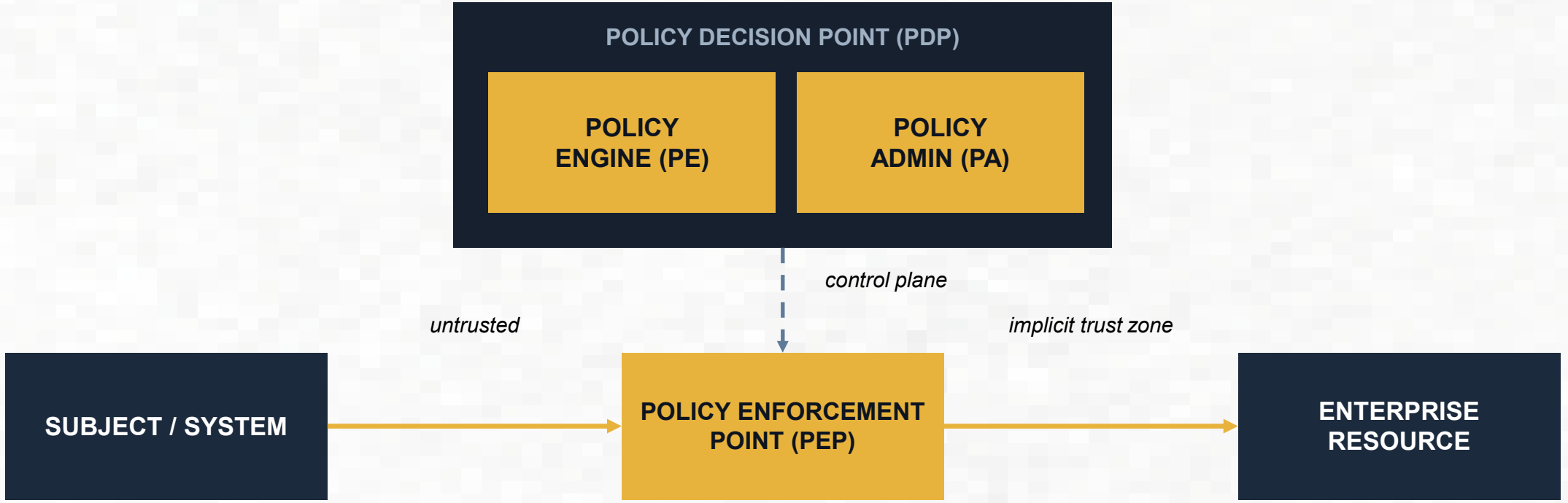
THE SEVEN TENETS OF ZERO TRUST

1. All data sources and computing services are resources
2. All communication is secured regardless of location
3. Access is granted on a per-session basis
4. Access is determined by dynamic policy (identity, device, behavior)
5. The enterprise monitors the integrity and posture of all assets
6. Authentication and authorization are dynamic and strictly enforced
7. Collect and use data to continually improve security posture

— NIST SP 800-207, Section 2.1

Source of Army Standards

LOGICAL COMPONENTS OF A ZTA



PE — decides. Grants, denies, or revokes access using enterprise policy plus inputs (CDM, threat intel, logs) fed to a trust algorithm.

PA — executes. Establishes or shuts down the communication path; issues session credentials and configures the PEP.

PEP — enforces. Enables, monitors, and terminates the connection between subject and resource. Nothing reaches a resource except through a PEP.

#StrongerArmyStrongerCountry

Source of Army Standards

DEPLOYMENT APPROACHES & VARIATIONS

Three approaches to building a ZTA (a full solution blends all three):

ENHANCED IDENTITY GOVERNANCE	Identity of actors is the key driver of policy. Access privileges of the subject decide; device and environment refine the confidence level. Suits open networks, BYOD, and cloud/SaaS.
MICRO-SEGMENTATION	Resources are placed on protected segments behind gateways (NGFWs, smart switches, host agents) acting as PEPs. Shields resources from unauthorized access and discovery.
SOFTWARE-DEFINED PERIMETER (SDP)	The network infrastructure itself implements ZT: the PA acts as network controller, reconfiguring paths per the PE's decisions (SDN / overlay networks).

Source of Army Standards

THREATS ASSOCIATED WITH ZTA

Subversion of the decision process

A compromised or misconfigured PE/PA can approve what it shouldn't. Configure, monitor, log, and audit all changes.

Denial of service / disruption

PE/PA/PEP become mission-critical choke points. Replicate them and host in resilient, secured environments.

Stolen credentials / insider threat

MFA and contextual trust algorithms detect abnormal access patterns and stop lateral movement.

Network visibility

Encrypted traffic can't all be inspected — analyze metadata and apply machine learning to spot malicious flows.

Storage of network information

Stored traffic, scans, and policies are reconnaissance gold. Protect them with the most restrictive policies.

Proprietary formats & NPEs

Vendor lock-in and automated agents (weaker authentication) introduce risk. Evaluate providers holistically.

Source of Army Standards

MIGRATING TO A ZTA — SEVEN STEPS

1. **Identify actors**
   Catalog subjects — users, admins, and non-person entities (service accounts).
2. **Identify assets**
   Inventory hardware, virtual assets, and shadow IT; monitor configuration and state.
3. **Identify key processes**
   Rank business processes and risks; start with a low-risk workflow.
4. **Formulate policies**
   Set criteria or confidence-level weights for the candidate workflow's resources.
5. **Identify solutions**
   Evaluate candidate tools against the workflow, assets, and environment.
6. **Deploy & monitor**
   Operate in reporting-only mode first; tune policies against real traffic.
7. **Expand the ZTA**
   Refine, then take on the next workflow. Reevaluate whenever the mission changes.

Source of Army Standards

APPLICABILITY TO PHILIPPINE ARMY

Secure tactical networks

Place PEPs in front of command networks and transmission systems; no resource is reachable without one.

Assume the network is hostile

Treat internal, deployed, and coalition networks as compromised; encrypt and authenticate everything.

Enable remote & coalition access

Grant partners and remote users per-session, least-privilege access without VPN-style broad trust.

Continuous monitoring

Feed logs, device posture, and threat intel to the policy engine — a core cyber-battalion function.

Contain the breach

ZTA prevents lateral movement, limiting the blast radius of stolen credentials or insider threats.

Commander's intent for cyber

Dynamic access policy is the digital expression of command decisions on acceptable mission risk.

SUMMARY

- ✓ Zero Trust: never trust, always verify — no implicit trust by location or ownership
- ✓ Zero Trust Framework Continually evaluate trust; enforce least-privilege, per-session access
- ✓ Seven tenets guide design; PE, PA, and PEP make and enforce every decision
- ✓ Three approaches: identity governance, micro-segmentation, SDP
- ✓ Threats remain — protect the PE/PA, plan for disruption, monitor continuously
- ✓ Migration is incremental: inventory, policy, pilot, tune, expand
- ✓ For the PA: ZTA secures tactical networks and contains the breach

FEEDBACK



FUTURE

Next lesson will be **CEOI / CESI Management**



THANK YOU!



HAVE A NICE DAY

180921 22H
BEAT BOUTIQUE